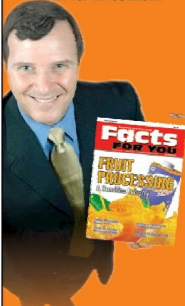


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Conventional LED casings are designed such that there is an air pocket within the lighting structure to contain heat, which makes the air pocket act like an oven. Greenstar decided to avoid this air gap in Galaxy XD casing

be mounted at any angle between 0 to 90 degrees. This has opened the usability of the casing not just for streetlights but also for building facade, sports stadium and parking lot lighting. The lights being modular in design, the same basic platform can be used to create nine different models.

Chandra confesses, "Creating a driver design where the lumens-per-watt efficacy remains constant while the light is modular was difficult to achieve. We created a modular casing and driver, which could cover a range of 45 watts to 175 watts lights. Other manufacturers offer different models for different wattages."

Galaxy XD can accommodate 24, 30, 36, 42, 48, 54, 60, 66 or 72 LEDs for differing levels of lumen output.

Moreover, conventional LED casings are designed such that there is an air pocket within the lighting structure to contain heat, which makes the air pocket act like an oven. Greenstar decided to avoid this air gap in Galaxy XD casing with a zero-air-gap 'unibody' construction. Any heat produced within the structure is immediately dissipated. Due to the space saved, the casing structure looks more compact, slim and slender.

New approach to sensors

Sensor-based streetlights have existed for a while, but Greenstar claims to have used a different approach. Chandra illustrates, "We use four semiconductor-based sensors in our lights, controlled by a microcontroller. These are configurable as per the specific requirements of the users. The microcontroller gives a higher degree of control on how the sensors work. As a result, functions like dimming can be precisely controlled. Besides, it

offers four levels of redundancy, which ensures that the sensors are always accurate, under all circumstances." Moreover, since the sensors are semiconductor devices, these are highly reliable. If any of the sensors fails, the others automatically provide a backup.

Software and integration

The software for the product was also developed by the engineers at Greenstar. It helps to customise the product to the user's needs precisely. Dr Chuck Chakravarty, COO, Sarosij Sengupta, VP, product development and Chandra took eight months to develop the platform for the product, and another two months to integrate it at the system level. Integration was the most challenging stage of the product development.

Chandra admits, "The stage where hardware and software come together was the most pressurising one. We worked very closely with our hardware complement suppliers. Our engineers travelled frequently to the component manufacturers' location and worked with their teams."

Looking ahead

Greenstar has an aggressive roadmap for its product development. It plans to introduce streetlights with communication capability, which will eventually put streetlights on a network. "This will enable one to monitor aspects like the health of the streetlights, performance, and energy saving statistics from a central remote location. We believe in continual product innovation and will have a series of path-breaking products in the coming months," Chandra concludes. ●

The author is from EFY Bureau, Bengaluru